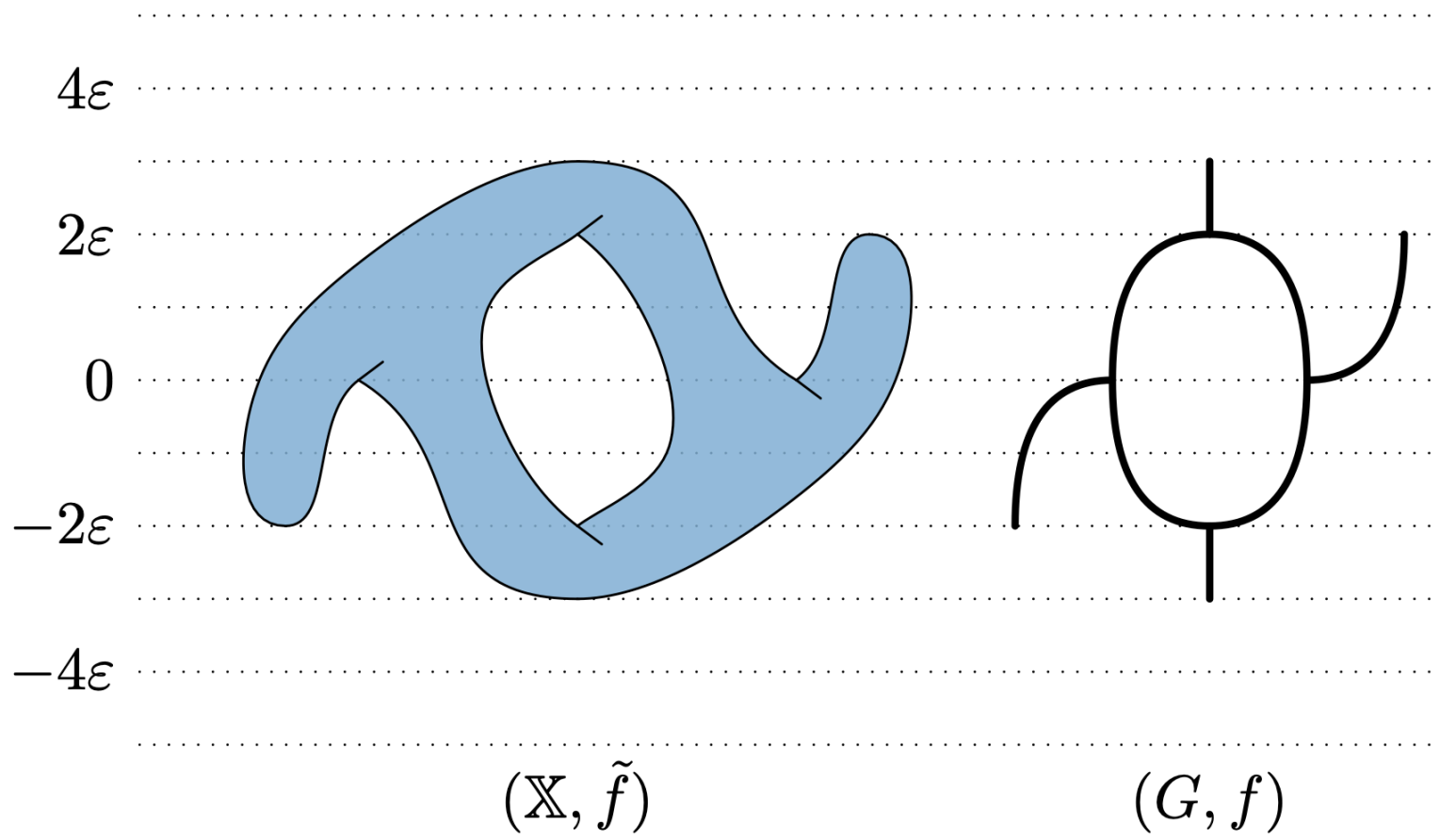


Presentation of: A family of metrics from the truncated smoothing of Reeb graphs

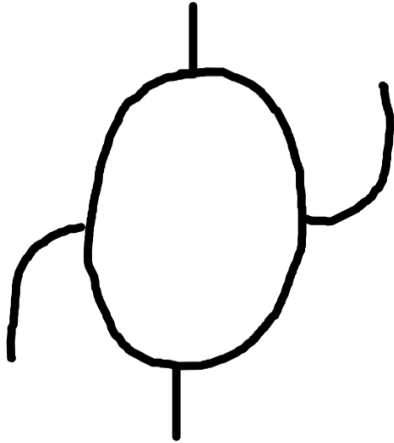
Kerzreho Yann, Hû Romain, Thomas Winninger¹

¹yann.kerzreho@ens-paris-saclay.fr, romainhu@laposte.net, thomas.winninger@telecom-sudparis.eu

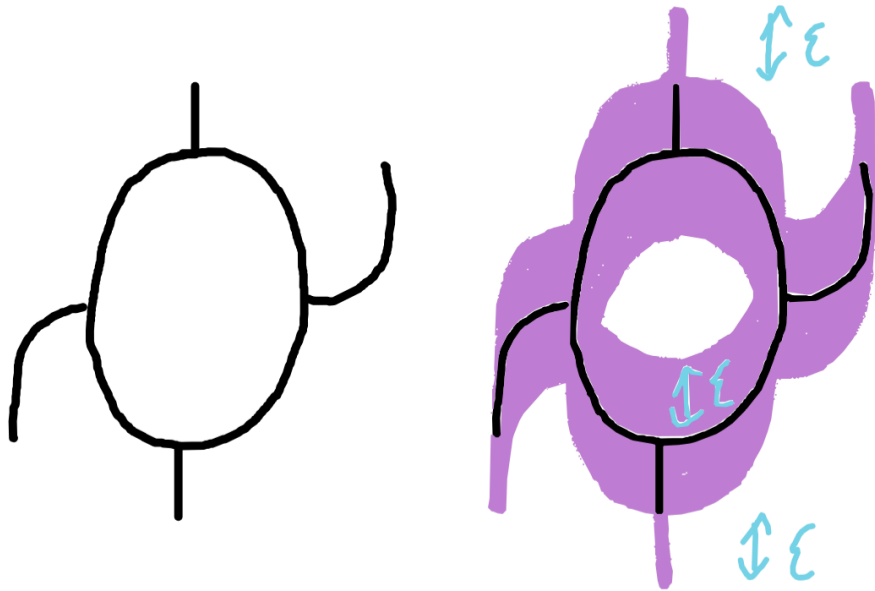


Presentation of: A family of metrics from the truncated smoothing of Reeb graphs

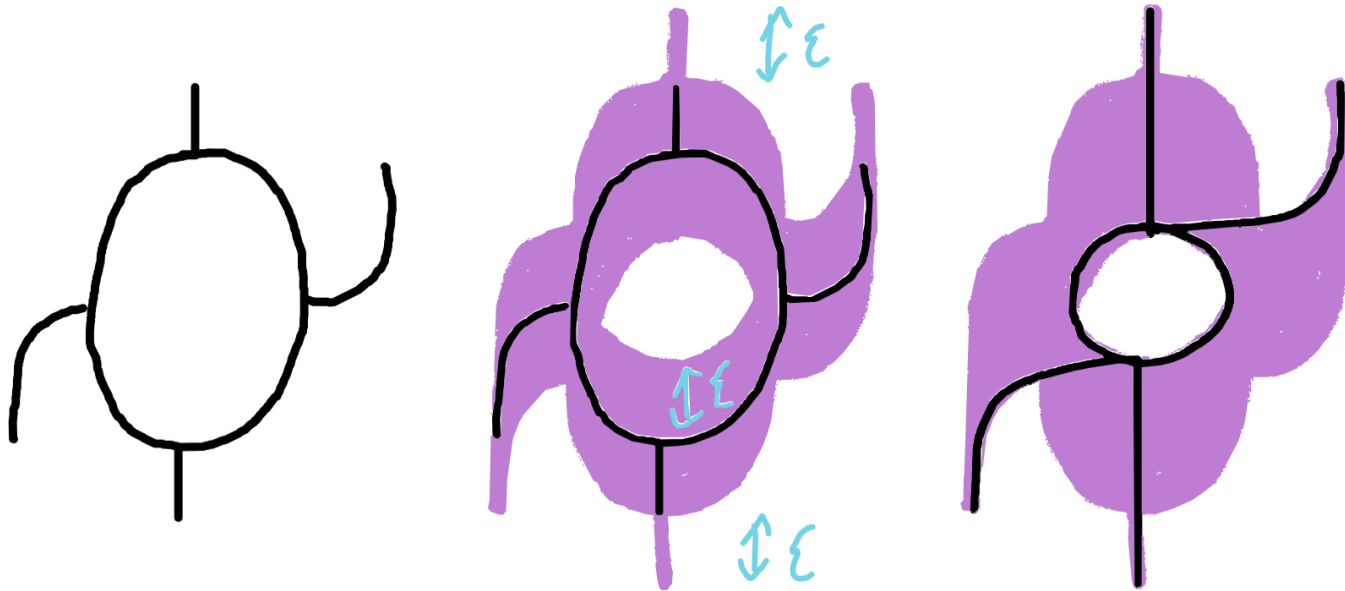
Smoothing a Reeb Graph



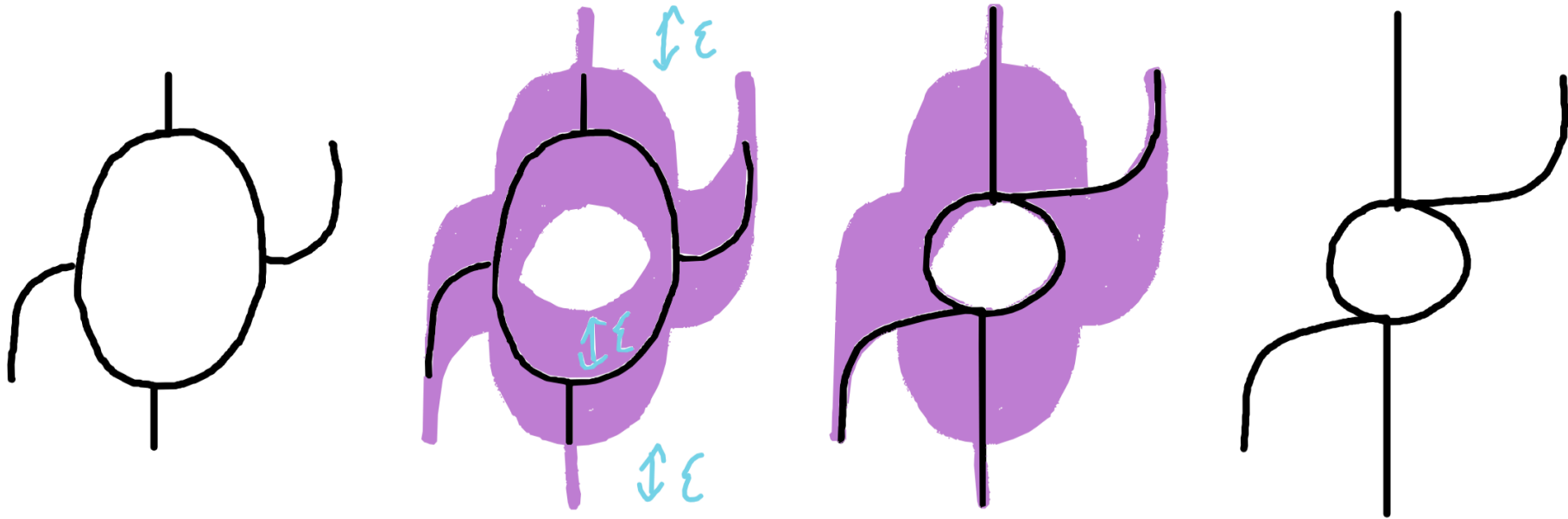
Smoothing a Reeb Graph



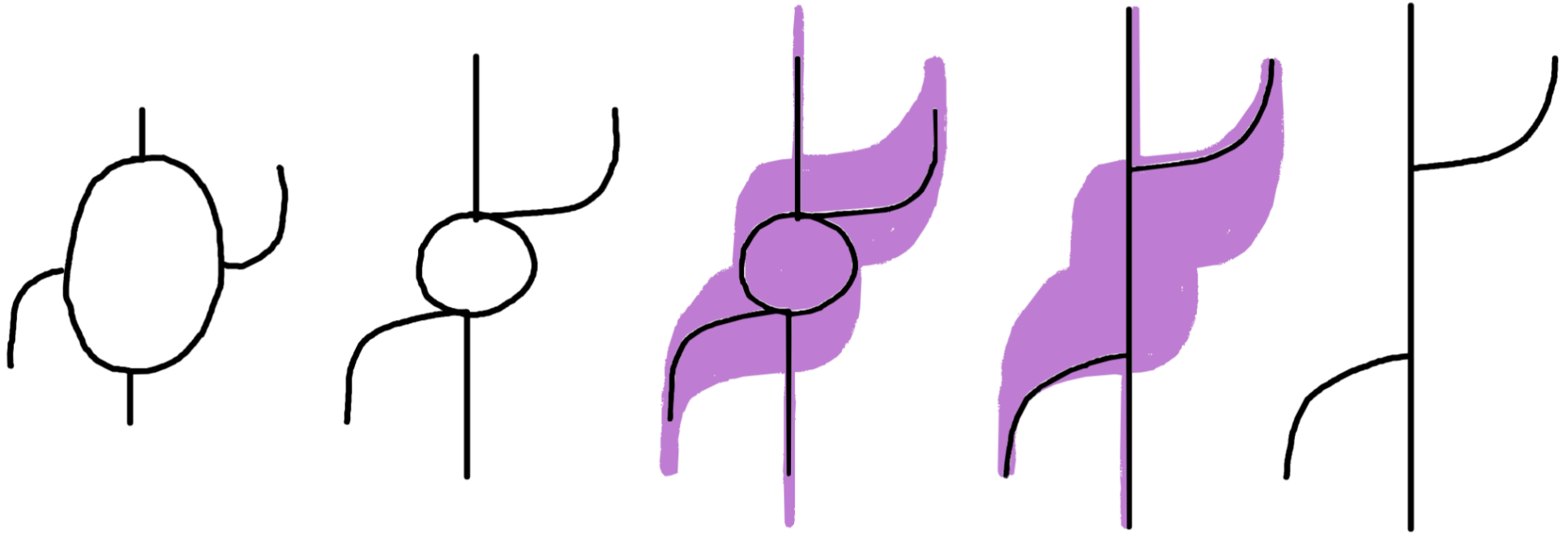
Smoothing a Reeb Graph



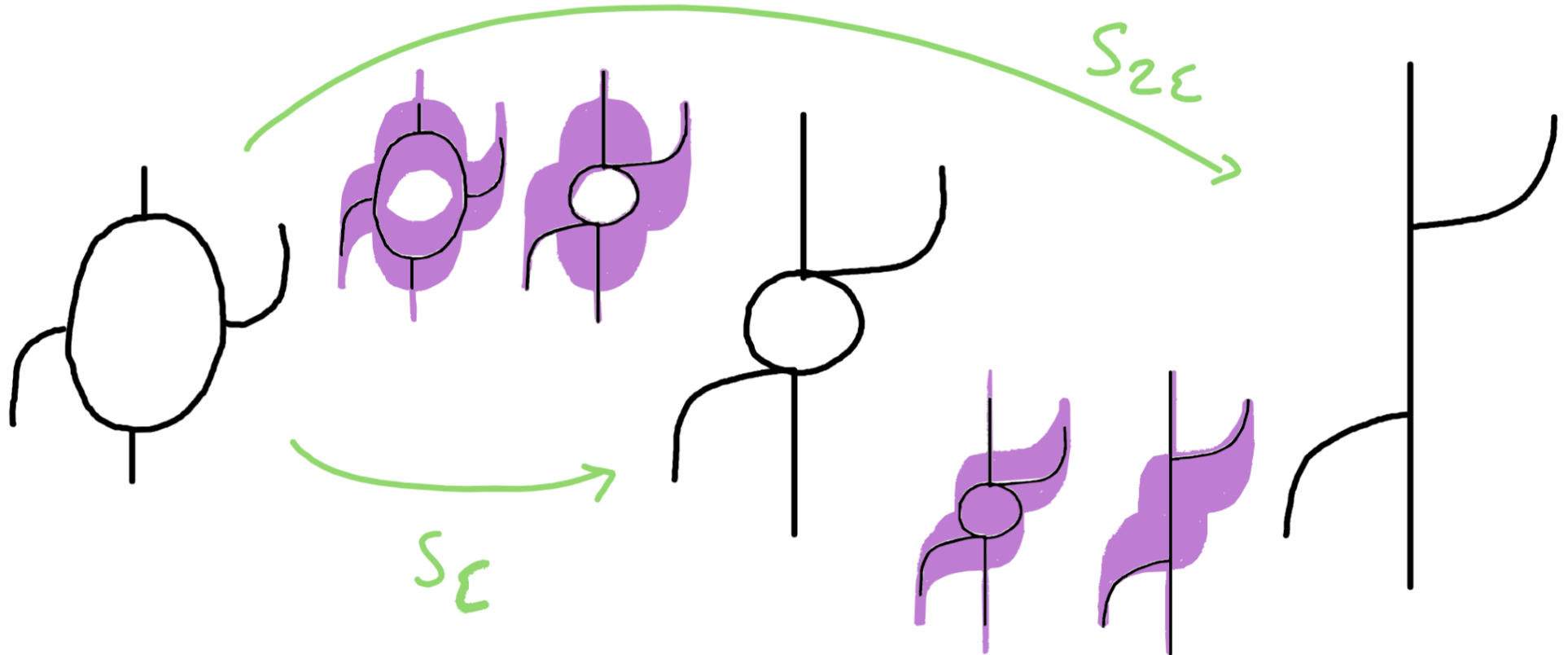
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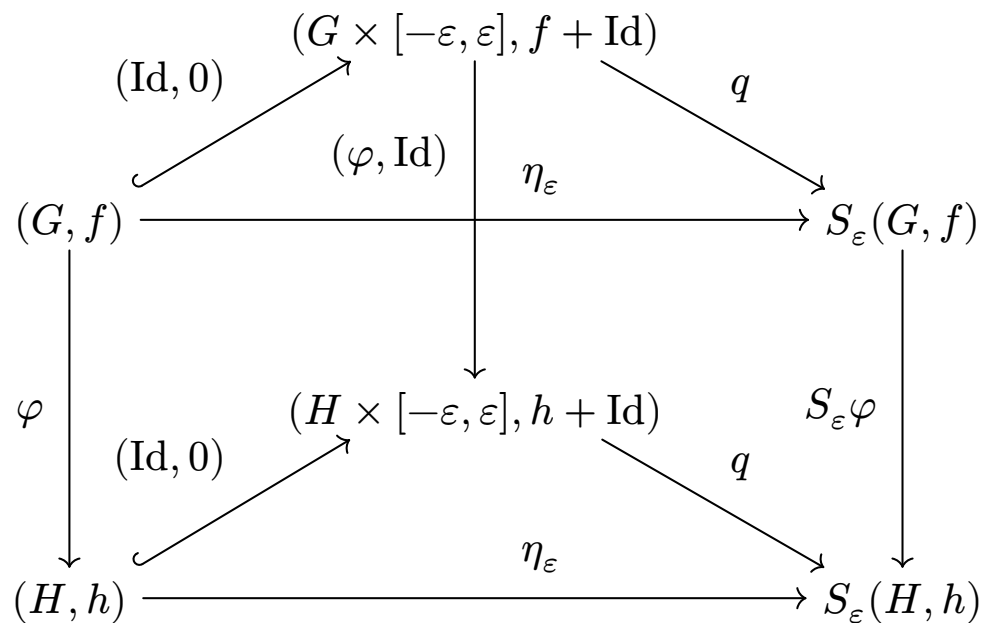
What is a Categorical Flow ?

Given a category $\mathcal{C}$, a flow is a functor $F : [0, \infty[\rightarrow \text{End}(\mathcal{C})$ with a semigroup structure: $F_0 \cong \mathbb{1}_{\mathcal{C}}$ and $F_{\varepsilon} F_{\varepsilon'} \cong F_{\varepsilon + \varepsilon'}$. In other words, the following diagram commutes:

$$\begin{array}{ccccc}
 (G, f) & \xrightarrow{F_{\varepsilon}} & F_{\varepsilon}(G, f) & \xrightarrow{F_{\varepsilon'}} & F_{\varepsilon + \varepsilon'}(G, f) \\
 \downarrow \varphi & & \downarrow F_{\varepsilon} \varphi & & \downarrow F_{\varepsilon + \varepsilon'} \varphi \\
 (H, h) & \xrightarrow{F_{\varepsilon}} & F_{\varepsilon}(H, h) & \xrightarrow{F_{\varepsilon'}} & F_{\varepsilon + \varepsilon'}(H, h)
 \end{array}$$

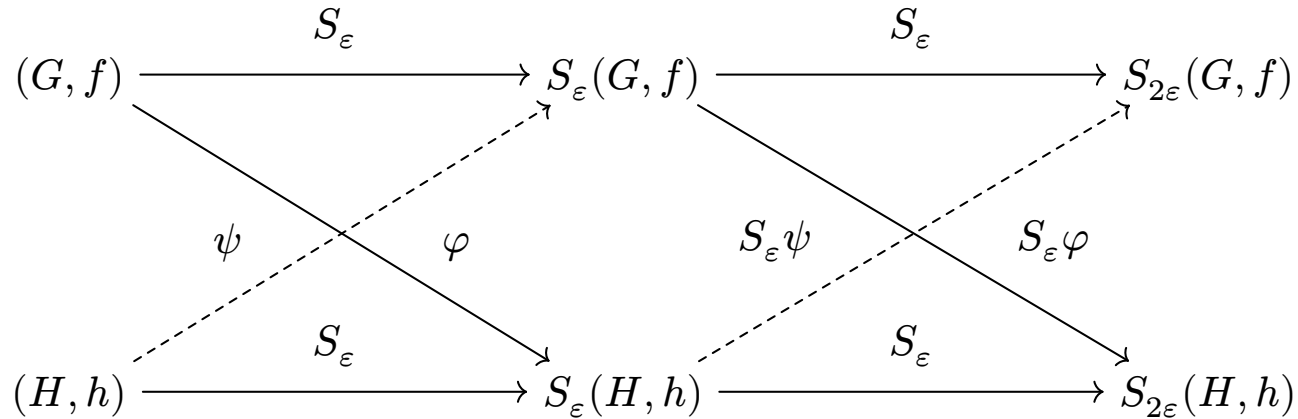
How to make a Smoothing Flow ?

We have a natural flow $\eta : ([0, \infty[, \leq) \rightarrow \text{End}(\mathcal{C}), \varepsilon \mapsto \eta_\varepsilon$ built with quotients:



Reminder

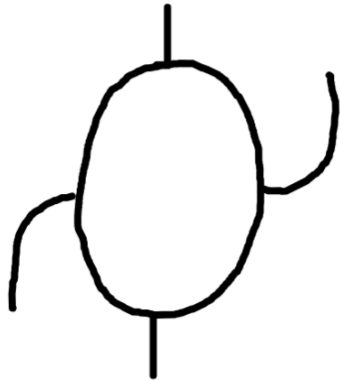
Given two Reeb graphs (G, f) , (H, h) , an ε -interleaving between (G, f) and (H, h) wrt S^m is a pair of maps $\varphi_\varepsilon, \psi_\varepsilon$ such that the following diagram commutes:



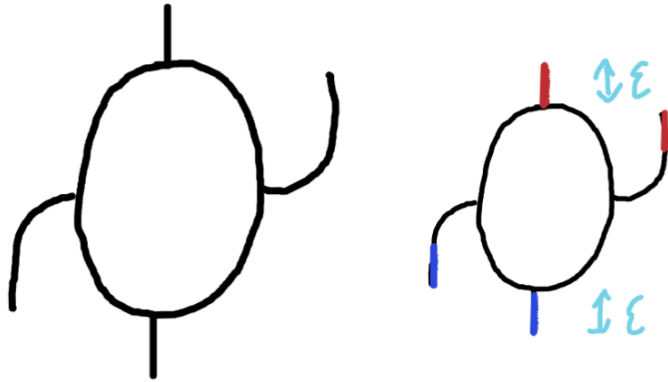
$$d_I((G, f), (H, h)) = \inf\{\varepsilon \geq 0 \text{ such that there exists an } \varepsilon\text{-interleaving wrt } S\}.$$

Extending with Truncated Smoothing

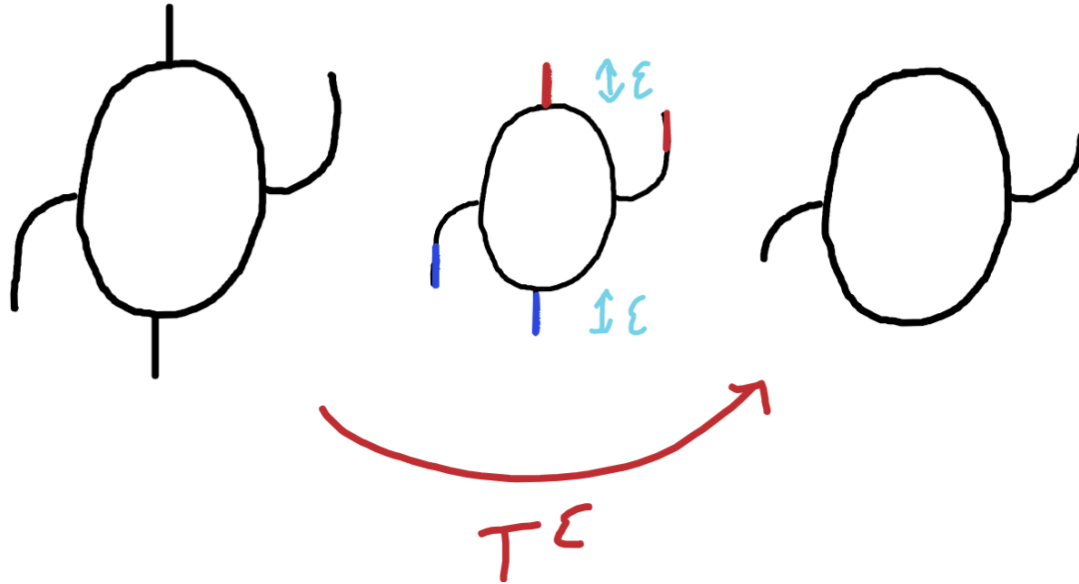
Truncating a Reeb Graph



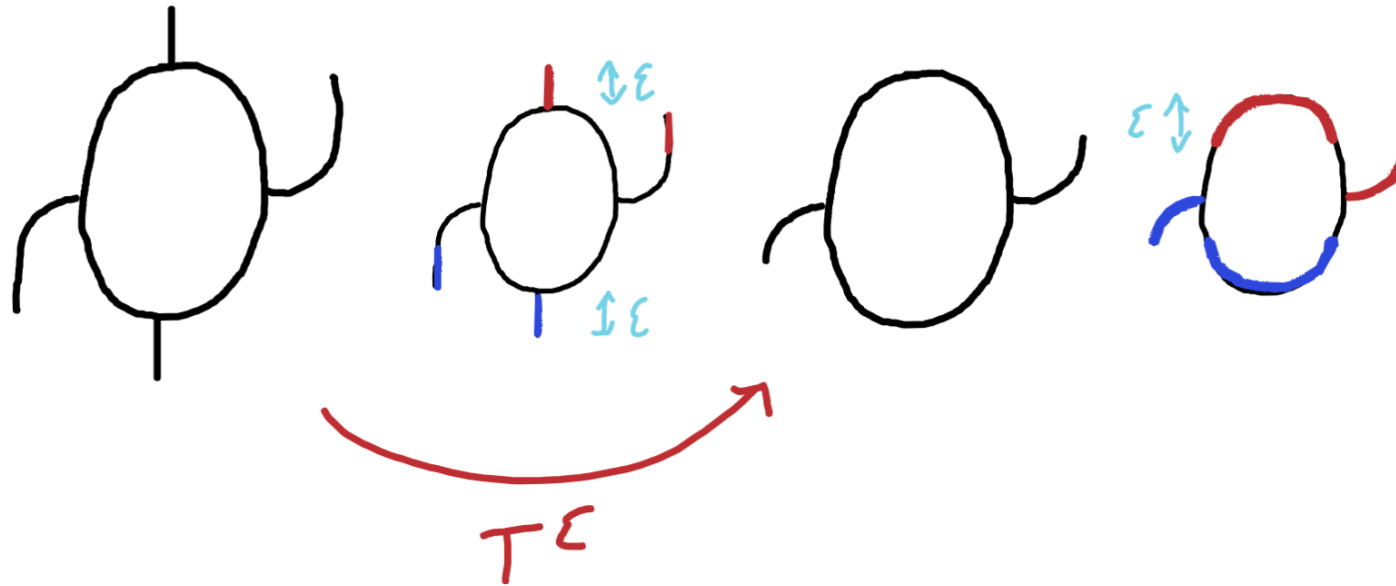
Truncating a Reeb Graph



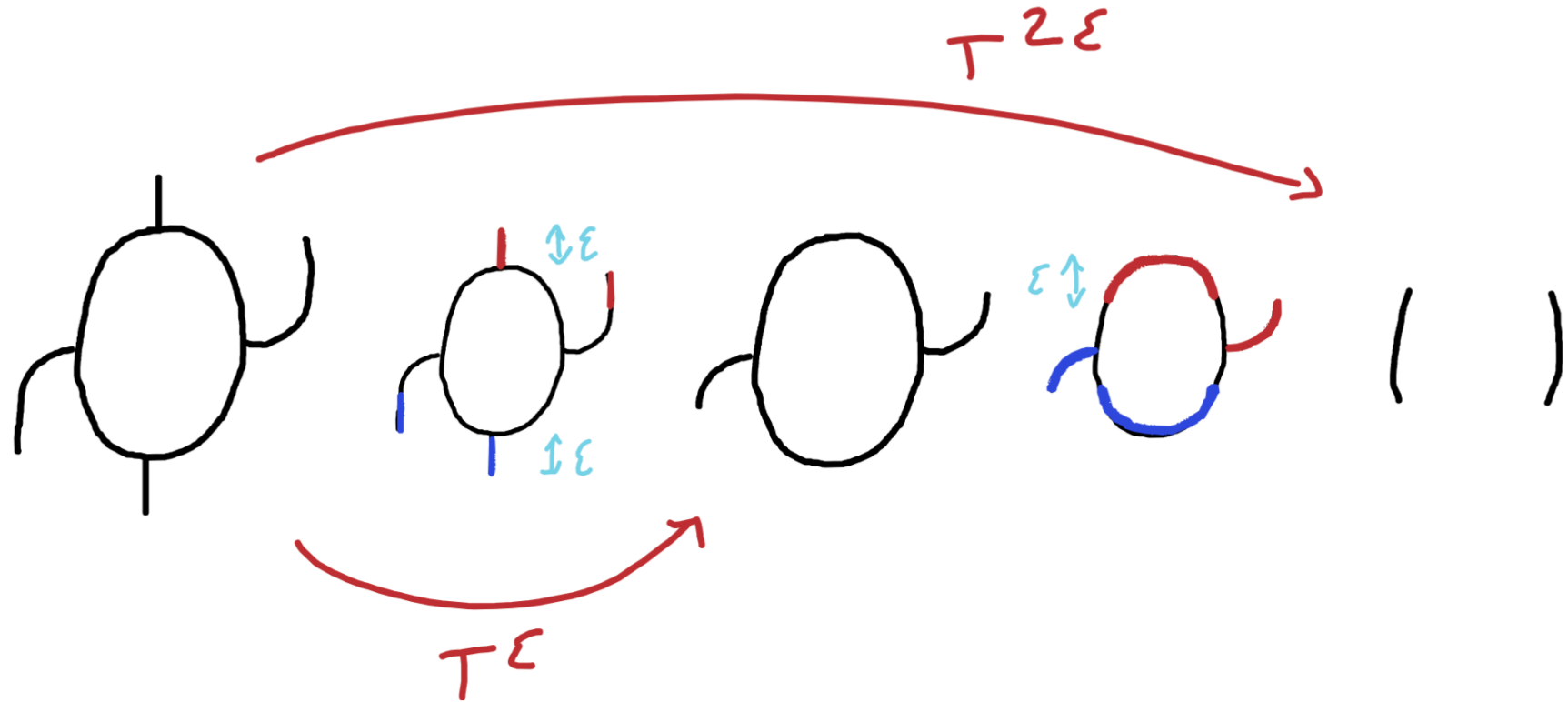
Truncating a Reeb Graph



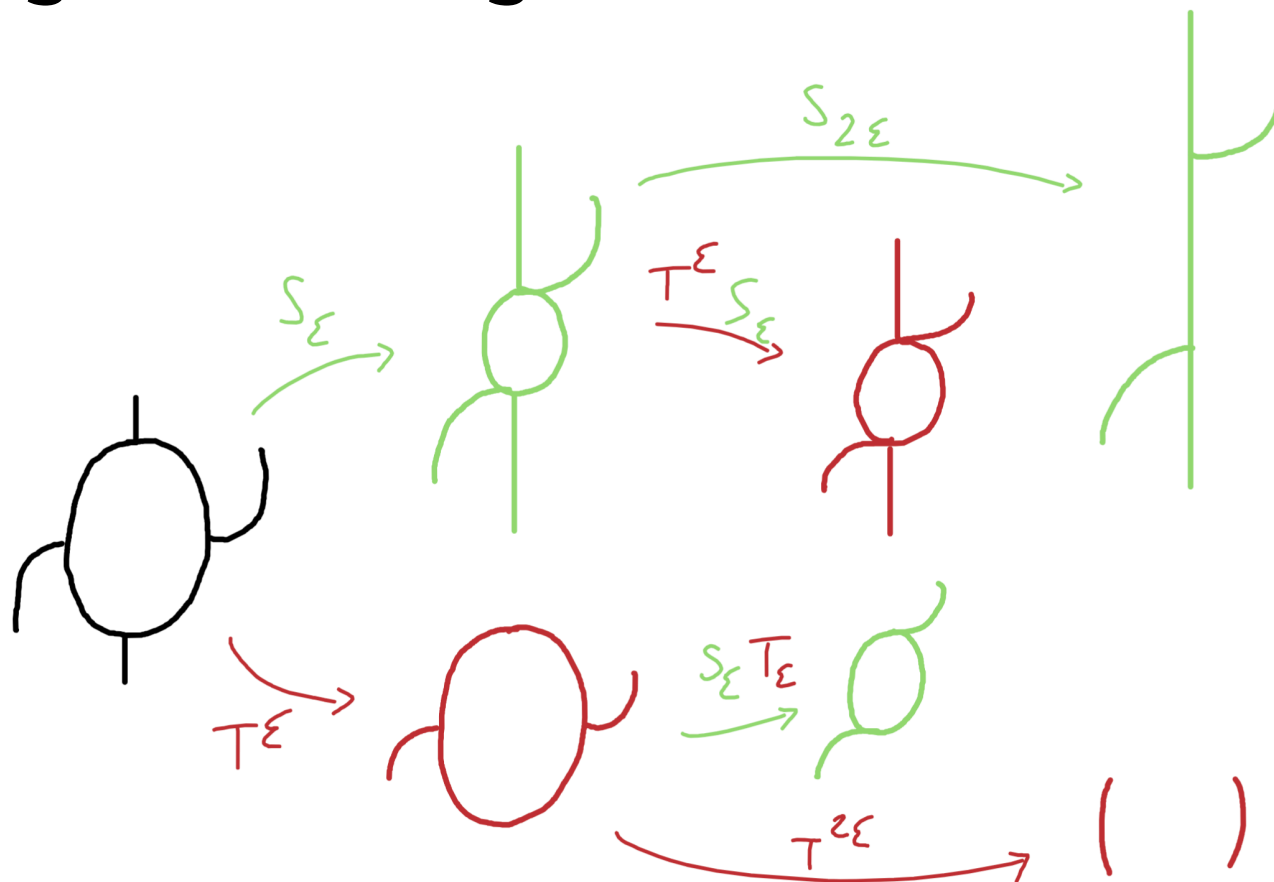
Truncating a Reeb Graph



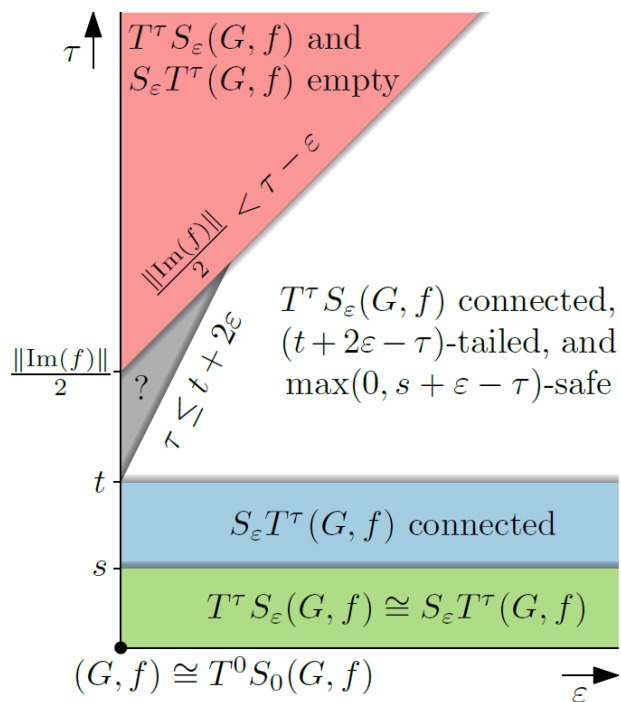
Truncating a Reeb Graph



Smoothing + Truncating



How to preserve properties under truncation ?



For t -tailed and s -safe graph:

- Empty if $\frac{\|Im(f)\|}{2} < \tau - \varepsilon$;
- Connected when $\tau < t$;
- T^τ and S_ε commute if $\tau < s$.

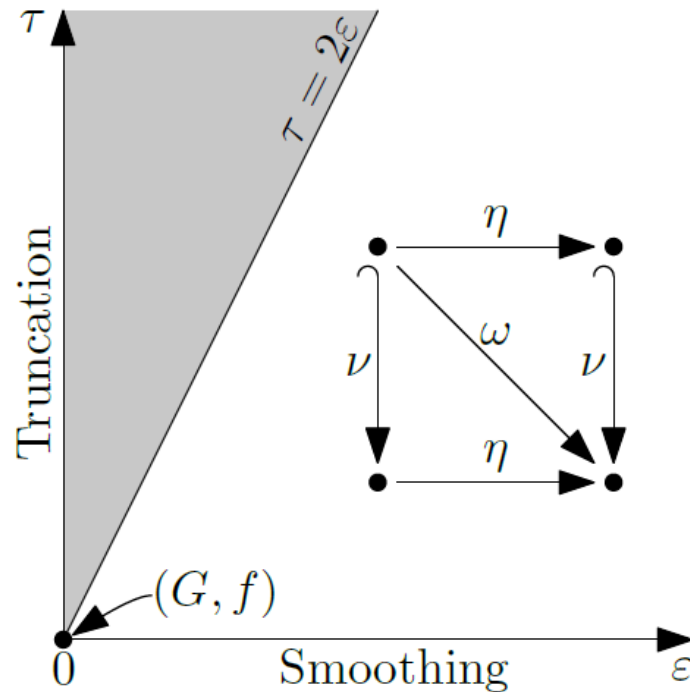
Truncated-Smoothing Interleaving

How to make a Truncation Flow ?

We have a natural “reverse” flow $\nu : ([0, \infty[, \geq) \rightarrow \text{End}(\mathcal{C}), \tau \mapsto \nu_{-\tau}$ built with inclusions:

$$\begin{array}{ccc}
 T^\tau(G, f) & \xrightarrow{\nu_{-\tau}} & (G, f) \\
 \downarrow \varphi & & \downarrow \nu_{-\tau}\varphi \\
 T^\tau(H, h) & \xrightarrow{\nu_{-\tau}} & (H, h)
 \end{array}$$

Combine Both

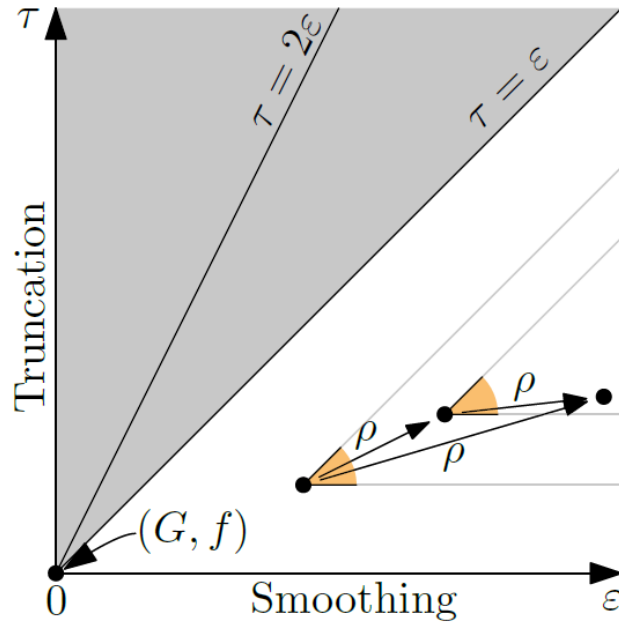


By combining η and ν , we obtain a flow with 2 generators:

$$\omega : (\varepsilon, \tau) \mapsto \omega_{\varepsilon}^{-\tau} = S_{\varepsilon}^{-\tau} = \nu_{-\tau} \eta_{\varepsilon} = \eta_{\varepsilon} \nu_{-\tau}$$

Problem: ω can only decrease truncation !

How to make a Flow with Increasing Truncation ?



It can be shown that $\text{Im}(\eta_\tau : S_{\epsilon-\tau}^0(G) \rightarrow S_\epsilon)(G) = S_\epsilon^\tau(G)$.
From there, we can build:

$$\begin{array}{ccccccc}
 G & \xrightarrow{\eta_{\epsilon-\tau}} & S_{\epsilon-\tau}^0(G) & \xrightarrow{\eta_\tau} & \text{Im}(\eta_\tau(S_{\epsilon-\tau}^0)) & \hookrightarrow & S_\epsilon \\
 & \searrow \rho & & \searrow \rho & & \searrow \rho & \\
 & & & & & & \\
 & \searrow \eta_\epsilon & & & & &
 \end{array}$$

Possible for $\tau \in [0, \epsilon] \Rightarrow$ for $m \in [0, 1]$, we have a flow
 $S^m : ([0, \infty[, \leq) \rightarrow \text{End}(\mathcal{C}), \epsilon \mapsto S_\epsilon^{m\epsilon}$.

Flow

Given two Reeb graphs (G, f) , (H, h) , an ε -interleaving between (G, f) and (H, h) wrt S^m is a pair of maps $\varphi_\varepsilon, \psi_\varepsilon$ such that the following diagram commutes:

$$\begin{array}{ccccc}
 (G, f) & \xrightarrow{S_\varepsilon^{m\varepsilon}} & S_\varepsilon^{m\varepsilon}(G, f) & \xrightarrow{S_\varepsilon^{m\varepsilon}} & S_{2\varepsilon}^{2m\varepsilon}(G, f) \\
 & \searrow \psi & \nearrow \varphi & \searrow S_\varepsilon^{m\varepsilon}\psi & \nearrow S_\varepsilon^{m\varepsilon}\varphi \\
 (H, h) & \xrightarrow{S_\varepsilon^{m\varepsilon}} & S_\varepsilon^{m\varepsilon}(H, h) & \xrightarrow{S_\varepsilon^{m\varepsilon}} & S_{2\varepsilon}^{2m\varepsilon}(H, h)
 \end{array}$$

$d_I^m((G, f), (H, h)) = \inf\{\varepsilon \geq 0 \text{ such that there exists an } \varepsilon\text{-interleaving wrt } S^m\}.$

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- Lipschitz-stability: $d_I^m \leq C \|f - h\|_\infty$;
- Strongly equivalent to distortion distance, Bottleneck distance $\leq 5d_I^m$.

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- Computing approximations of interleaving distances ;
- Finding optimal bounds for strong equivalence ;
- Prove convergence of algorithms ;
- Improve resistance to noise ;
- Truncated-smoothing preserves planarity of Reeb graphs ;
- Unify the ever growing family of Reeb graph metrics.